Jacob Chalk

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Research Focus

My research focus is on audio-visual video understanding, particularly in egocentric videos. Here, my aim is to leverage both audio and video modalities to their maximum potential for solving challenging problems such as action recognition and detection. More recently, I have also begun exploring video understanding in 3D.

Education

PhD in Computer Vision *September 2021 – Current*
University of Bristol *Bristol, UK*

- *Thesis Title: Audio-Visual Video Understanding*

MEng in Computer Science *September 2017 – September 2021*
University of Bristol *Bristol, UK*

- *Degree Classification: First Class Honours*
- *Dissertation Title: Video GANS for Human-Object Interactions (83%)*

Teaching

Teaching Assistant *September 2019 – Current*
University of Bristol *Bristol, UK*
Software Product Engineering (Y2), Games Project (Y3), Computer Graphics (Y3) Image Processing and Computer Vision (Y3), Applied Deep Learning (Y4)

Honours and Awards

EgoVis 2022/23 Distinguished Paper Awards *June 2024*
First Joint Egocentric Vision Workshop *IEEE/CVF Conference (CVPR)*

EPIC-KITCHENS Challenges Winner *June 2024*
First Joint Egocentric Vision Workshop *IEEE/CVF Conference (CVPR)*
Audio-Based Interaction Recognition (2nd), Audio-Based Interaction Detection (2nd), Action Detection (3rd)

Top 5 Third Year MEng Computer Science/Computer Science with Maths *October 2020*
Netcraft *University of Bristol*

Publications

- *Chalk, J., Huh, J., Kazakos, E., Zisserman, A. and Damen, D., 2024. TIM: A Time Interval Machine for Audio-Visual Action Recognition. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 18153-18163)*
- *Huh, J., Chalk, J., Kazakos, E., Damen, D. and Zisserman, A., 2023, June. Epic-sounds: A large-scale dataset of actions that sound. In ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.*

Research Activities

Presentations

- EPIC-KITCHENS Challenges. Presented at the First Joint Egocentric Vision Workshop (CVPR 2024).
- Oral Presentation for EPIC-Sounds. Presented at ICASSP 2023.

Reviewing

- Reviewer for ECCV 2024 and TPAMI (2024)